MEGN 651A: Homework Five

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The code used to produce this work can be located at this address:

https://github.com/rachbertaud/Advanced_CFD/tree/main/05_membrane_channel_mass_transfer.

For reproduction of results, download the folder `05_membrane_channel_mass_transfer` and enter the directory of the corresponding case you would like to reproduce. Simply run `$ make build` and `$ make run` to produce the main data files. For code verification, enter the `Verf` directory and run `$ make build` then `$ make ErrorData`. Then, run `errorplot.m`. For all other cases, simply run `viz.m` in the respective folder to produce and save the plots provided below. `Main` also contains a fun `$ make DiffData` call to produce diffusion data, along with `diffplot.m` to visualize results.

Here are the following case types:

- `Verf` - Verifies the updated structs described in problems 6.1 - 6.3.
- `Unstable` - Runs the main code with Neumann BC on the east wall, which causes oscillations.
- `Main` - Runs the physical example of a membrane with extrapolation condition on the east boundary.

`$ make clean` removes all the executables and all the `txt` files in the respective folders.

Overview of Code

The purpose of this code is to numerically solve steady, two-dimensional advection-diffusion equation* at discrete locations using a finite-volume method with physical additions to the boundaries. Our goal is to simulate the concentration distribution of dissolved or suspended species over a membrane, specifically interested in concentration polarization, where the species accumulate near the membrane, making filtration difficult.

We consider a two-dimensional channel of length L_x and height L_y. The channel walls are semi-permeable membranes through which liquid passes with a constant transmembrane velocity V_m. We consider the idealized case of a liquid containing a single dissolved or suspended species, which we refer to as the solute. The fluid velocity field is approximated using the analytical solution below,

$$u(\hat{x},\hat{y}) = \frac{3}{2}U_{in}\left(1-\frac{V_m}{U_{in}}\hat{x}\right)\left(1-\hat{y}^2\right),$$

$$v(\hat{y}) = \frac{V_m}{2}\left(3\hat{y}-\hat{y}^3\right),$$

where

$$\hat{x} = \frac{x}{h}, \qquad \hat{y} = \frac{y-h}{h},$$

and $h = L_y/2$ is the channel half-height.

Gov. Equations

The solute transport is governed by the steady advection-diffusion equation,

$$u\frac{\partial c}{\partial x} + v\frac{\partial c}{\partial y} = D\left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2}\right),$$

where $c(x, y, t)$ is the solute concentration and D is the mass diffusivity. We consider the case where the system is initialized with zero solute concentration everywhere,

$$c\Big|_{t=0} = 0.$$

At time $t = 0$, a uniform concentration C_{in} is applied at the inlet,

$$c\Big|_{x=0} = C_{in}, \quad t > 0.$$

At the membrane walls, we assume the solute cannot pass through the membrane. This requires the sum of convective and diffusive transport of solutes in the y-direction to sum to zero at the membrane surfaces,

$$\left[-D\frac{\partial c}{\partial y} + vc\right]_{y=0,\,L_y} = 0,$$

where $v = -V_m$ at $y = 0$ and $v = V_m$ at $y = L_y$. We originally define the conditions

$$c\Big|_{x=0} = C_{in}, \quad \text{where} \quad C_{in} = 1.$$

and

$$\frac{\partial c}{\partial x}\Big|_{x=L_x} = 0.$$

This is the `Unstable` case. For our `Main` case, we perform an extrapolation boundary when $x = L_x$, where

$$c_{N+1} = c_N + \frac{c_N - c_{N-1}}{x_N - x_{N-1}}(x_{N+1} - x_N).$$

Verification of Main

We first define our manufactured solution for the verification steps as

$$T_e(x, y) = \sin(x)\sin(y).$$

We discretize as defined for the steady, 2D advection-diffusion equation* with boundary conditions for each section that fits the form

$$aT + b\frac{\partial T}{\partial n} = g.$$

These are quick computations and will not be defined here.

We build $\mathbf{A}$ and $\mathbf{b} = C_e$ (and fitting our BC) as explicitly defined in the overview section, and solve for $\mathbf{c}$. Using the results of this step, we verify that

1. Our solution's relative error measured with the infinity norm converges with order two accuracy and
2. Our operator, $\mathbf{A}$, computes the expected solution for our governing equations.

The results of the verification process are defined below in the "Results" section.

Error Analysis

During the verification process, error of the numerical solution is calculated as relative error using the infinity norm, which is defined as

$$E_\infty = \frac{||c_{\text{numerical}} - c_{\text{analytical}}||_\infty}{||c_{\text{analytical}}||_\infty} = \frac{\max|c_{\text{numerical}} - c_{\text{analytical}}|}{\max|c_{\text{analytical}}|}.$$

Note that we will also define local error as

$$\text{Local Error} = |c_{\text{numerical}} - c_{\text{analytical}}|.$$

Results

In the problems that follow, we fix

$$U_{in} = 0.1 \ \mathbf{m/s}, \quad V_m = 10^{-5} \ \mathbf{m/s}, \quad L_x = 0.05 \ \mathbf{m} \quad \text{and} \quad L_y = 0.002 \ \mathbf{m}.$$

Overall, the results of this code verified its capabilities and produced an intuitive solution to our little room. Everything performed as anticipated. Refer to

- Figure 1 for results of `Verf` — where we confirmed second-order on our new Robin conditions.
- Figure 6 for results of `Unstable` — where we investigated Neumann conditions on the east boundary.
- Figure **??** for results of `Main` — where we added extrapolation to the east boundary to remove wiggles.

Problem 6.4 (Verification Results)

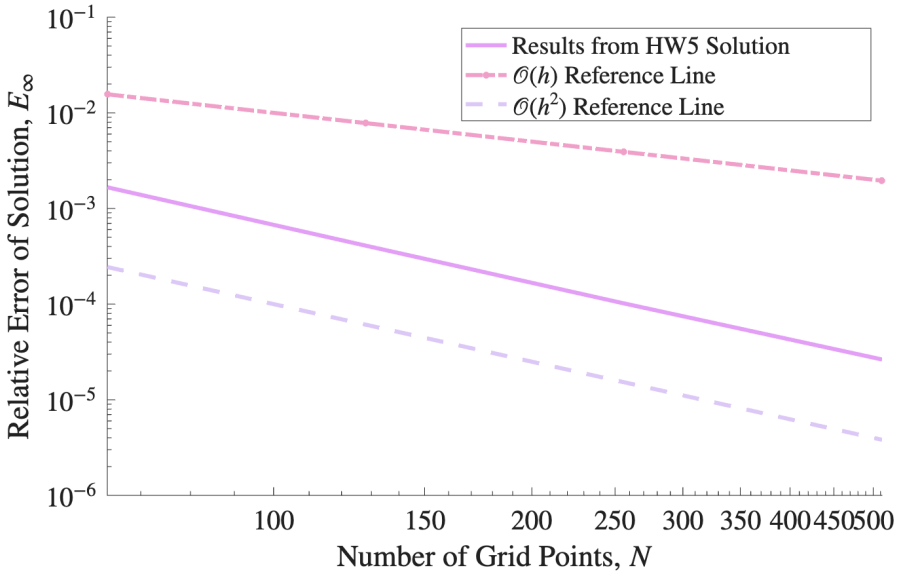
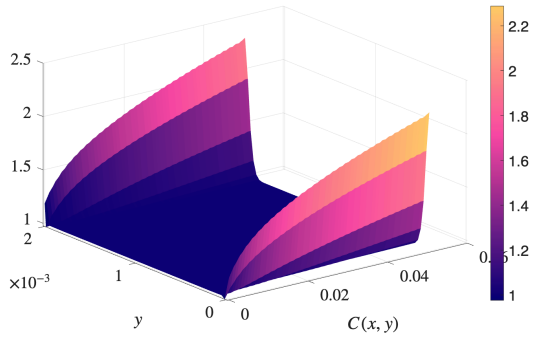
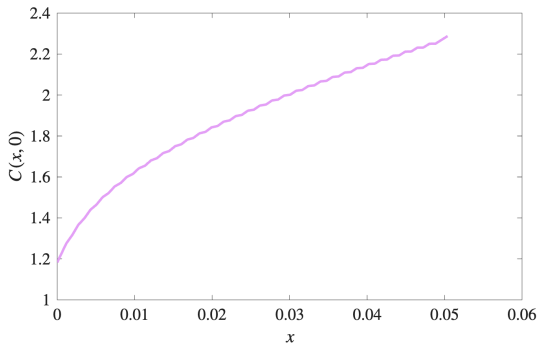


Figure 1: Relative ℓ_∞ error E_∞ of the numerical solution as a function of the number of grid points N, compared against $\mathcal{O}(h)$ and $\mathcal{O}(h^2)$ reference lines. The solution converges at second-order, consistent with the $\mathcal{O}(h^2)$ reference.

Problem 6.5 (Post-Processing)

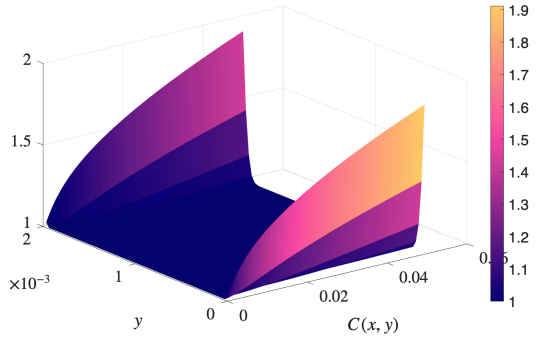


(a) 3D surface plot of $c(x, y)$ over full domain.

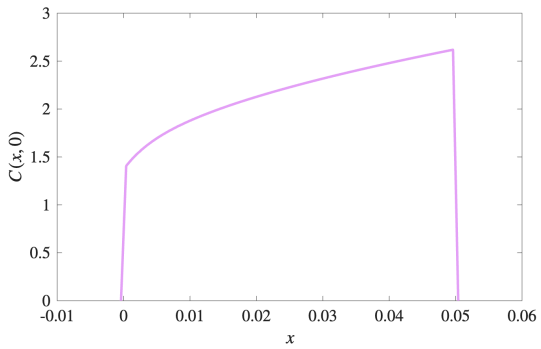


(b) Plot of $c(x, 0)$ as a function on x on the lower membrane ($y = 0$).

Figure 2: Numerical concentration results for unstable case, where the fluctuations (wiggles) on the edges are apparent. Note for this case that $Nx = Ny = 64$ and $\mathcal{D} = 1\text{e} - 9.\text{m}$ **Note the wiggles are independent of mesh resolution.**

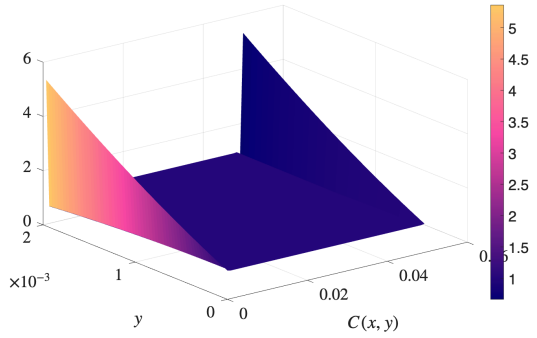


(a) 3D surface plot of $c(x, y)$ over full domain.

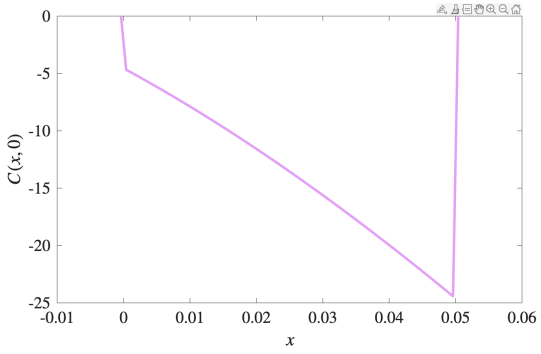


(b) Plot of $c(x, 0)$ as a function on x on the lower membrane ($y = 0$).

Figure 3: Numerical concentration results for main case. Note for this case that $Nx = Ny = 64$ and $\mathcal{D} = 1\text{e} - 9.\text{m}$

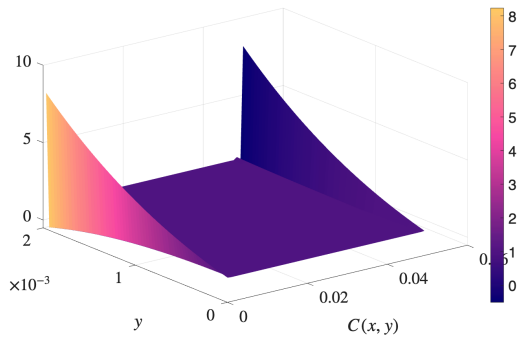


(a) 3D surface plot of $c(x, y)$ over full domain.

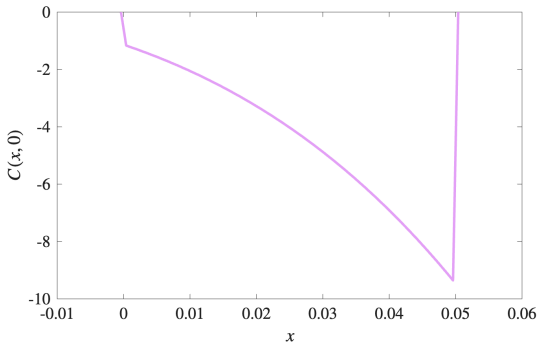


(b) Plot of $c(x, 0)$ as a function on x on the lower membrane ($y = 0$).

Figure 4: Numerical concentration results for main case. Note for this case that $Nx = Ny = 64$ and $\mathcal{D} = 1\text{e} - 10.\text{m}$

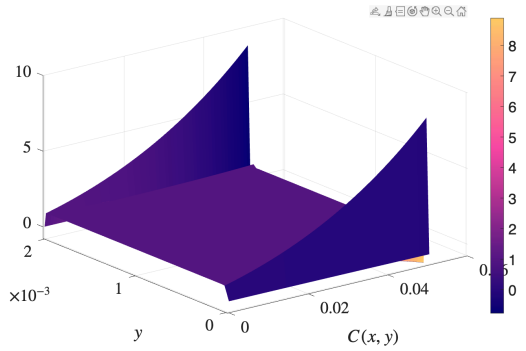


(a) 3D surface plot of $c(x, y)$ over full domain.

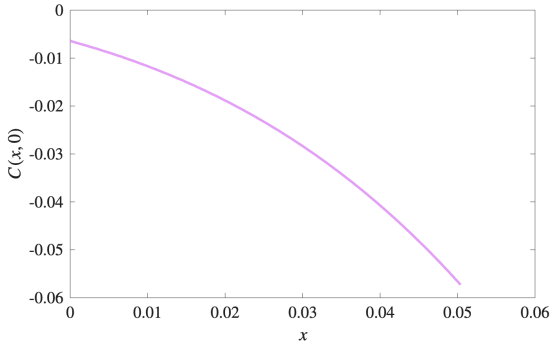


(b) Plot of $c(x, 0)$ as a function on x on the lower membrane ($y = 0$).

Figure 5: Numerical concentration results for main case. Note for this case that $Nx = Ny = 64$ and $\mathcal{D} = 1\text{e} - 11.\text{m}$



(a) 3D surface plot of $c(x, y)$ over full domain.



(b) Plot of $c(x, 0)$ as a function on x on the lower membrane ($y = 0$).

Figure 6: Numerical concentration results for main case. Note for this case that $Nx = Ny = 64$ and $\mathcal{D} = 1\text{e} - 12.\text{m}$

Problem 6.7 (Effect of Diffusivity on Concentration Polarization)

Below are the answers to the questions asked for in the HW assingment:

1. My results are semi-sensitive to mesh resolution. For the $\mathcal{D} = 1e-10$ case, I was able to remove the boundary instabilities seen above by increasing my grid resolution. As $\mathcal{D}$ became smaller, the edges no longer responded to reasonable increases in grid resolution.

2. As diffusion decreases, I believe the thickness of the concentration layer increase near the ends of the membrane and decrease near the beginning. It is hard to see from these plots.

3. As the diffusion decreases, the maximum concentration at the membrane surface increases, which corresponds to the following question. This is clear from the plots.

4. The results have the distribution along the membrane as a direct result of advection and diffusion. Diffusion disperses the particles are every point, and advection carries them along the velocity lines. As we investigate the decrease of diffusion, we would expect more particles following the streamlines then diffusing around, which is why (more visually than from the (b) plots) that the concentration actually becomes exponential from beginning to the end, suggesting more particles are ending up at the end of the membrane. When there is diffusion, we get more particles at the beginning where they are less likely to end up with advection alone based on our velocity definitions.

Code

For the sake of page usage, I am only including the main code. Everything else that was used to produce these results can be found at this GitHub location. Please refer to the first page for reproduction instructions.

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main.cpp

```
#include "Header_VolSF.H"
#include "Header_Mesh.H"
#include "Header_velF.H"
#include "Header_ScaSF.H"
#include "Header_matrix.H"
#include "Header_bcType.H"

int main()
{
  int flag = 0;

  meshType Mesh;

  volScalarField C("C", Mesh);

  velocityField U(Mesh);

  surfaceScalarField Flux(Mesh);
  Flux.ComputePhix(Mesh, C, U);
  Flux.ComputePhiy(Mesh, C, U);

  matrixType Matrix(Mesh);
  Matrix.BuildA(Mesh, Flux, C);
  Matrix.BuildRHS(Mesh, C, U);

  Matrix.EigenSolve();



  return flag;
};
```